

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT – 4 – FALL SEMESTER 2026-2027

Programme Name & Branch: M.Tech. (Integrated) CSE (Data Science)

Course Name: Data Science Programming Lab

Course Code: CSI3004

Assessment 4:

Tweet and Retweet Data Analysis Using R

Case Study: Identifying the Spread of Fake News on Social Media

A social-media analytics company wants to investigate how fake and genuine news propagate through social media. The company has collected tweet-level information containing the tweet ID, user, date, tweet text, number of retweets, likes, replies, and a `Fake/Real` label.

Using the R programming language, analyze the dataset and identify the characteristics and spread pattern of fake-labelled information.

Perform tweet and retweet data analysis using R to identify the spread pattern of fake news.

You are provided with a dataset containing the following attributes:

Attribute	Description
<code>Tweet_ID</code>	Unique identifier of the tweet
<code>User</code>	User who posted the tweet
<code>Date</code>	Date of posting
<code>Text</code>	Tweet content
<code>Retweets</code>	Number of retweets
<code>Likes</code>	Number of likes
<code>Replies</code>	Number of replies
<code>Label</code>	Fake or Real

Tasks

1. Data Creation and Exploration – 10 Marks

- Create/load the dataset in R.
- Display the first 10 records.
- Display the structure and summary statistics.
- Determine the number of Fake and Real tweets.

2. Retweet Analysis

- Calculate the total number of retweets for Fake and Real tweets.
- Calculate the average number of retweets for each category.
- Identify the tweet with the maximum number of retweets.
- Determine whether Fake or Real tweets have higher average retweet activity.

3. Engagement Analysis

- Calculate the engagement score for every tweet.
- Calculate the average engagement for Fake and Real tweets.
- Identify the top five tweets based on engagement.

4. Temporal Spread Analysis

- Calculate the total number of retweets for Fake and Real tweets for each date.
- Plot a line graph showing the spread of Fake and Real tweets over time.
- Identify the date on which Fake-labelled tweets achieved their highest propagation.
- Explain the observed spread pattern.

5. Influential User Analysis

Calculate the following for each user:

- Number of tweets
- Total retweets
- Average retweets
- Total engagement

Identify the top three users based on total retweets.

6. Visualization

Create at least two appropriate visualizations, such as:

- Fake vs. Real tweet distribution
- Retweet comparison
- Engagement comparison
- Daily spread pattern
- Top viral tweets

Provide suitable titles, axis labels and legends.